

Question ID: 9f2ecade

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$h(x) = x^3 + ax^2 + bx + c$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are -5 , 6, and 7, what is the value of c ?

Rationale

The correct answer is 210. Since -5 , 6, and 7 are zeros of the function, the function can be rewritten as $h(x) = (x + 5)(x - 6)(x - 7)$. Expanding the function yields $h(x) = x^3 - 8x^2 - 23x + 210$. Thus, $a = -8$, $b = -23$, and $c = 210$. Therefore, the value of c is 210.